



Cox Proportional Hazards Modelling of Time-to-Event Outcomes Among HIV Patients on Antiretroviral Therapy at Specialist Hospital Yola, Adamawa State, Nigeria

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Abstract: Traditional cross-sectional methods could not adequately capture time-to-event dynamics among HIV patients on antiretroviral therapy (ART). Survival analysis provides a more appropriate framework for examining predictors of clinical outcomes. This study applied a multivariable Cox proportional hazards model to examine socio-demographic and clinical factors associated with time to treatment interruption or death among 2,887 HIV patients on ART at Specialist Hospital Yola, Adamawa State, Nigeria. The outcome was time to treatment interruption or death, with Current ART Status serving as the failure indicator (1 = interruption in treatment (IIT) or dead; 0 = censored). Independent variables were age group, sex, marital status, and viral load status (suppressed <1000 copies/ml versus unsuppressed ≥1000 copies/ml). Kaplan-Meier curves, Log-rank tests, and Cox regression models were employed in the investigation. The proportional hazards assumption was assessed using log-minus-log plots, Schoenfeld residuals, and the global Schoenfeld test. Most participants were females (70.1%) and had suppressed viral load (95%). Unsuppressed viral load was significantly associated with treatment interruption or death in both univariate (hazard ratio (HR) = 1.98; 95% confidence interval (CI): 1.44–2.73; $p < 0.001$) and multivariate analyses (adjusted HR = 1.96; 95% CI: 1.42–2.70; $p < 0.001$). Age group, sex, and marital status were not significant predictors. The proportional hazards assumption was satisfied (global Schoenfeld test $p = 0.669$). Unsuppressed viral load was therefore the only independent predictor of treatment interruption or death. Strengthening viral load monitoring and adherence support should be prioritized to improve treatment outcomes.

Keywords: Antiretroviral therapy; Cox proportional hazards model; HIV; Survival analysis; Viral load

1. Introduction

The evaluation of HIV and AIDS has shifted from simple case counting to the modelling of complex and time-dependent clinical outcomes (Luvanda et al., 2023). Globally, the expansion of highly active antiretroviral therapy (HAART) has driven public health systems toward longitudinal patient tracking (Montaner et al., 2014). While traditional epidemiological indicators such as prevalence provide a useful cross-sectional snapshot of the proportion of infected individuals at a given point in time, they do not capture temporal variations in disease progression, patient retention, or the occurrence of clinical events (World Health Organization, 2023). As a result, modern biostatistics increasingly relies on survival analysis techniques, particularly the Cox proportional hazards regression model, to examine the relationship between risk factors and the time until specific clinical events occur (Dzinza & Ngwira, 2022).

In Adamawa State, surveillance records from the Adamawa State Agency for the Control of AIDS (ADSACA) indicate that the prevalence of HIV declined from 2.5% in 2014 to 1.1% (Leadership News, 2024). As of 2023, 36,137 people living with HIV were reported to be on antiretroviral medication in Adamawa State (SolaceBase, 2023).

Historically, most epidemiological research in Adamawa State relied on static methods such as log-linear

models, frequency counts, and binary logistic regression. For example, Akinrefon et al. (2023) applied log-linear models to examine associations among gender, age, viral load, and marital status among 3,779 HIV/AIDS patients attending Specialist Hospital Yola, between 2019 and 2020. While such methods are useful for identifying demographic associations with HIV status, they are limited when the research interest involves long-term clinical outcomes and time-to-event data (Dzinza & Ngwira, 2022). Traditional regression approaches typically collapse time-dependent outcomes into simple binary categories, thereby ignoring the temporal dimension of patient care and the presence of censorship arising from treatment default, relocation, or study closure (Dzinza & Ngwira, 2022).

As standard prevalence metrics and cross-sectional analyses do not account for time-to-event dynamics, public health administrators cannot accurately determine when patients face the highest risk of adverse outcomes. This methodological gap also limits the ability of clinicians to quantify how clinical markers, such as viral load, interact with demographic characteristics over time to influence patient trajectories (Dzinza & Ngwira, 2022).

1.1 Empirical Review

In a neighbouring North-Eastern state, Ahmadu et al. (2024) applied the Cox proportional hazards model to HIV patient records from Federal Medical Centre Jalingo, Taraba State. Their analysis identified CD4 count stage and marital status as significant predictors of the hazard of adverse outcomes, with patients in advanced clinical stages (stage IV) having approximately four times the risk of the event compared with those in earlier stages, and single patients having twice the risk of married patients. These findings from a contiguous state reinforce the value of survival modelling for understanding time-to-event dynamics among people living with HIV in the region and provide a useful comparative context for the present study at Specialist Hospital Yola.

Apeagee et al. (2020) applied the Cox proportional hazards model to time-to-event data of HIV/AIDS patients in Nigeria. Their study identified sex, age, and patients' health status as significant predictors of mortality, while higher CD4 cell counts were associated with a reduced risk of death. The findings provide further methodological support for the use of the Cox model in analysing survival outcomes among HIV patients in Nigerian treatment settings.

In the local setting, Akinrefon et al. (2023) used log-linear models to analyze HIV/AIDS data from 3,779 patients at Specialist Hospital Yola. Although the study successfully identified associations involving gender, age, viral load, and marital status, its reliance on static categorical analysis underscored the need for time-to-event methods tailored to the same facility and population.

Ikpe et al. (2025) conducted a retrospective cohort analysis of the Nigeria national HIV data repository to identify predictors of time to first treatment interruption among patients on antiretroviral therapy (ART) across 2,226 public facilities. Using Kaplan–Meier survival estimates, log-rank tests, and multivariable Cox proportional hazards regression, the study found that age grouping, ART regimen class, recorded viral load, CD4 cell count, World Health Organization (WHO) clinical stage, and facility location were predictive of treatment interruption. These findings underscored the value of national-level survival modelling for ART programme evaluation in Nigeria and provided a useful benchmark for facility-specific investigations such as the present study.

Dzinza & Ngwira (2022) compared parametric and Cox regression models using HIV survival data from Malawi. Their findings demonstrated the advantage of the Cox model in handling censored observations in resource-limited African settings, providing methodological support for its application in similar contexts such as Adamawa State.

Beyond West and Southern Africa, Chimnet et al. (2024) examined time to viral load suppression among HIV patients receiving ART in the Gebi Resu zone, Afar Region, Ethiopia. Applying Cox proportional hazards regression, the authors confirmed that the proportional hazards assumption was satisfied and identified baseline WHO clinical stage, cotrimoxazole prophylaxis therapy, and drug adherence as significant predictors of delayed viral suppression. This study reinforced the broader methodological validity of Cox-based survival modelling for monitoring ART outcomes in resource-limited settings across diverse African contexts.

Collectively, these studies showed that while static models remained common in parts of Nigeria, survival analysis techniques, particularly the Cox proportional hazards model, offer superior insight into temporal risk factors. However, there remains a notable scarcity of Cox-based survival studies specific to Adamawa State. The present study addressed this gap by applying a multivariable Cox proportional hazards regression model to examine the influence of socio-demographic and clinical factors on time to treatment interruption or death among HIV patients on antiretroviral therapy at Specialist Hospital Yola, Adamawa State, Nigeria.

2. Methods

2.1 Study Area

This study was conducted using HIV/AIDS patient data obtained from Specialist Hospital Yola, Adamawa State, Nigeria. The hospital is a major referral health facility in the state and provides comprehensive healthcare services,

including HIV/AIDS diagnosis, treatment, monitoring, and ART management. It serves patients from different parts of Adamawa State and the neighbouring regions, rendering these areas a core centre for HIV/AIDS care and clinical research.

2.2 Source of Data

The data used in this study were obtained from the medical records of HIV patients on antiretroviral therapy at Specialist Hospital Yola, Adamawa State. These records formed the basis of an earlier analysis using log-linear models that examined associations between socio-demographic factors and HIV/AIDS prevalence (Akinrefon et al., 2023). The present study re-analyzed the same facility-based dataset with the addition of follow-up time and event status coding for the application of Cox proportional hazards models.

2.3 Study Design

A retrospective survival analysis design was adopted using time-to-event data obtained from HIV/AIDS patient records. Survival analysis was considered appropriate because the study focused on the time until the occurrence of treatment interruption or death among patients on ART. The Cox proportional hazards regression model was applied to evaluate the effects of demographic and clinical factors on the survival outcome.

2.4 Study Population

The analysis included 2,887 HIV patients on ART who had documented follow-up time and outcome information. Patients with incomplete data on the key variables required for the survival analysis (age, sex, marital status, and survival time/event status) were excluded. Two participants had missing or invalid viral load values; they were retained in analyses that did not involve viral load and were excluded only from models and tables that included viral load status. Consequently, the Cox regression models that included viral load were based on 2,885 participants.

2.5 Variables in the Study

The dependent variable was survival time, defined as the duration in months from the date of ART initiation to the date of the recorded Current ART Status (or last clinical contact). The primary outcome was a composite event of interruption in treatment (IIT) or death. Current ART Status was coded as the failure indicators as follows: Event (failure = 1) = IIT or Dead; Censored (0) = Active, Active-Restart, Active-Transfer In, or Transferred Out.

The independent variables examined were age group (categorized as ≤ 29 years, 30–39 years, 40–49 years, and ≥ 50 years), sex (female or male), marital status (married, single, widowed, divorced, or separated), and viral load status. Viral load was classified as suppressed (< 1000 copies/ml) or unsuppressed (≥ 1000 copies/ml) according to the World Health Organization threshold.

2.6 Study Period and Data Extraction

The data used in this study were obtained from the medical records of HIV patients receiving antiretroviral therapy at Specialist Hospital Yola, Adamawa State. These records formed the basis of an earlier analysis using log-linear models that examined associations between socio-demographic factors and HIV/AIDS prevalence (Akinrefon et al., 2023). Permission to access the anonymized patient records was granted by the management of Specialist Hospital Yola for academic research purposes.

2.7 Cox Proportional Hazards Regression Model

The Cox Proportional Hazards Regression Model was used to assess the effects of the independent variables on the hazard of treatment interruption or death. The model proposed by Cox (1972) is given as:

$$h(t/X) = h_0(t)e^{(\beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4)} \quad (1)$$

where, $h(t/X)$ is the hazard function at time t given the covariate vector X , $h_0(t)$ is the baseline hazard function, and β is the vector of regression coefficients. In the present analysis, the covariate vector X contained indicator variables for age group (three indicators), sex (one indicator), viral load status (one indicator), and marital status (four indicators), giving a total of nine coefficients that were estimated.

2.8 Hazard Ratio

The hazard ratio (HR) was used to interpret the effect of explanatory variables on survival outcomes and is defined as:

$$HR = e^{\beta} \quad (2)$$

where, $HR > 1$ indicates an increased hazard (risk); $HR < 1$ indicates a reduced hazard (protective effect); $HR = 1$ indicates no effect.

2.9 Parameter Estimation

The regression parameters of the Cox model were estimated using the partial likelihood estimation method. The partial likelihood function is expressed as:

$$L(\beta) = \prod_{i=1}^n \left[\frac{e^{\beta X_i}}{\sum_{j \in R(t)} e^{\beta X_j}} \right] \quad (3)$$

where, the coefficients were obtained by maximizing this partial likelihood function.

2.10 Statistical Analysis

Data cleaning, coding, and analysis were carried out using Stata statistical software. Descriptive statistics were first used to summarize the baseline characteristics of the study participants. Categorical variables were presented as frequencies and percentages, while continuous variables were summarized using means, standard deviations, and medians. Differences between participants who experienced treatment interruption or death and those who did not were examined using the chi-square test.

Survival probabilities were estimated using the Kaplan-Meier method, and differences in survival experience across groups were compared using the Log-rank test. Univariate Cox proportional hazards regression was performed to obtain crude HRs and 95% confidence intervals. All variables were subsequently entered into a multivariate Cox model to estimate adjusted HRs while controlling for potential confounding.

The proportional hazards assumption was assessed both graphically and statistically. Log-minus-log plots and Schoenfeld residual plots were explored for the covariates included in the final model. In addition, the global Schoenfeld residual test was conducted. A non-significant result $p > 0.05$ was interpreted as evidence that the proportional hazards assumption was reasonably satisfied. A p -value of less than 0.05 was considered statistically significant.

3. Results and Discussion

3.1 Assessment of the Proportional Hazards Assumption

Figure 1 presents the Schoenfeld residual plots for the covariates in the final multivariate model. The plots showed no strong systematic trend over time. The smoothed residual lines remained roughly horizontal and close to zero. These graphical findings, together with the non-significant global Schoenfeld test ($p = 0.669$), confirmed that the proportional hazards assumption was reasonably satisfied.

Figure 2 shows the log-minus-log plots for viral load status, sex, age group, and marital status. For viral load status, the curves were approximately parallel, hence supporting the proportional hazards assumption. Plots for sex and age group also showed roughly parallel lines. Although some deviation was noted in the smaller marital status categories, the overall pattern did not indicate major violation of the proportional hazard assumption. These graphical findings are consistent with the formal Schoenfeld residual test.

3.2 Kaplan-Meier Survival Analysis

The Kaplan-Meier survival curve (Figure 3) stratified by viral load status revealed a clear separation between the two groups. Participants with unsuppressed viral load (≥ 1000 copies/ml) had consistently lower survival probability throughout the follow-up period than those with suppressed viral load (< 1000 copies/ml). This difference was highly statistically significant (Log-rank test, $p < 0.001$), indicating a strong association between unsuppressed viral load and increased risk of treatment interruption or death.

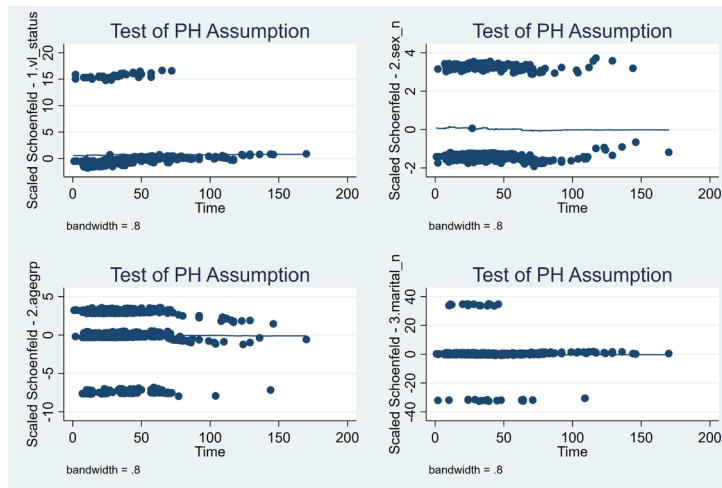


Figure 1. Schoenfeld residual plots for the covariates

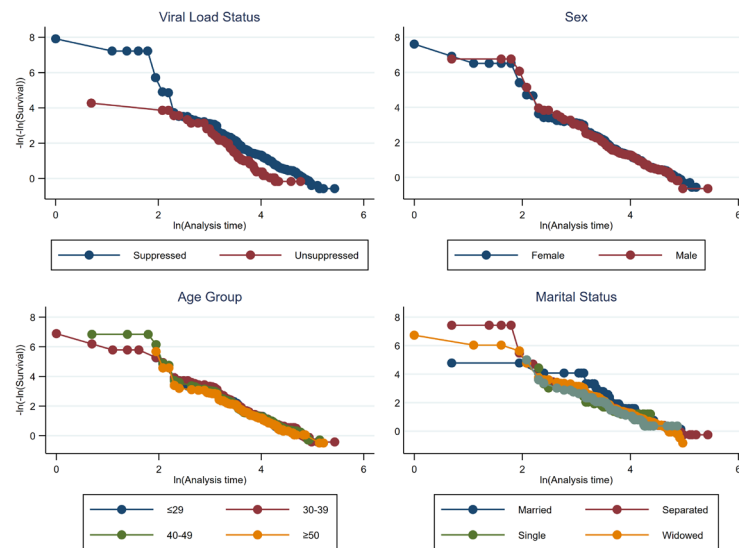


Figure 2. Log-minus-log plots

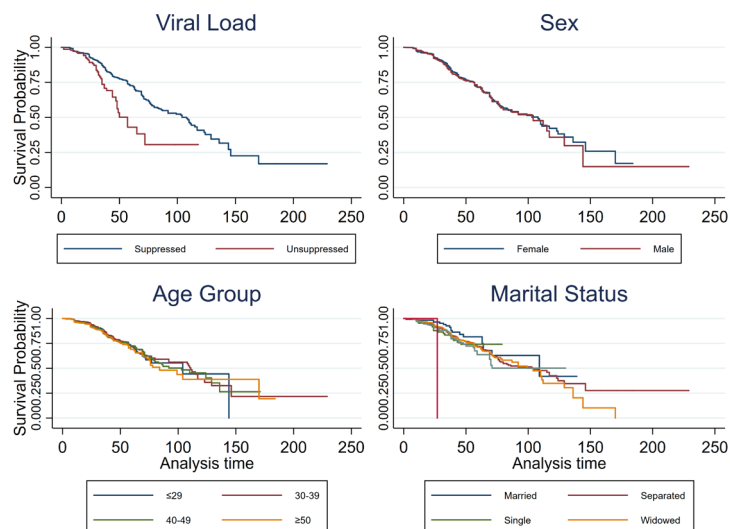


Figure 3. Kaplan-Meier survival curves

In contrast, the Kaplan-Meier curves by sex showed almost complete overlap between males and females across the entire observation period. The Log-rank test confirmed that there was no significant difference in survival experience between the two sexes ($p = 0.609$).

Similarly, when survival was examined according to age group (≤ 29 , 30–39, 40–49, and ≥ 50 years), the curves remained close to one another with no distinct pattern of separation. The difference across age groups was not statistically significant (Log-rank test, $p = 0.635$), suggesting that age did not substantially influence time to treatment interruption or death in this cohort.

The Kaplan-Meier curves by marital status displayed minor variations, particularly among the smaller categories (widowed, divorced, and separated). Although the overall Log-rank test reached statistical significance ($p = 0.028$), the visual separation between the major groups (married and single) was limited, indicating that the practical difference in survival by marital status was modest. Note that the chi-square tests in Table 1 assess the cross-sectional association between each covariate and the occurrence of treatment interruption or death, whereas the Log-rank tests evaluate differences across the entire survival curves.

Table 1. Baseline socio-demographic and clinical characteristics of participants ($N = 2,887$)

Characteristic		Total (%) ($N = 2,887$)	No Event (%) ($n = 2,276$)	Event (%) ($n = 611$)
Sex	Female	2,023 (70.1)	1,601 (70.3)	422 (69.1)
	Male	864 (29.9)	675 (29.7)	189 (30.9)
Age group	≤ 29 years	386 (13.4)	302 (13.3)	84 (13.8)
	30–39 years	979 (33.9)	781 (34.3)	198 (32.5)
	40–49 years	934 (32.4)	731 (32.1)	203 (33.3)
	≥ 50 years	587 (20.3)	462 (20.3)	125 (20.5)
Marital status	Married	1,687 (58.4)	1,330 (58.4)	357 (58.4)
	Single	839 (29.1)	662 (29.1)	177 (29.0)
	Widowed	154 (5.3)	115 (5.1)	39 (6.4)
	Divorced	120 (4.2)	101 (4.4)	19 (3.1)
Viral load status	Separated	87 (3.0)	68 (3.0)	19 (3.1)
	Suppressed (<1000 copies/ml)	2,740 (95.0)	2,170 (95.4)	570 (93.3)
	Unsuppressed (≥ 1000 copies/ml)	145 (5.0)	104 (4.6)	41 (6.7)
Follow-up time (units)	Median	36	37	35
	Mean $\pm$ SD	40.8 $\pm$ 23.7	41.3 $\pm$ 23.4	39.1 $\pm$ 24.7

Note: Two participants had missing or invalid viral load values and were excluded only from analyses and the viral load row that involved viral load status; the total analytical sample for models including viral load was $N = 2,885$. One participant had missing age information and was excluded from age-group rows. The follow-up time was reported in months. SD = standard deviation.

3.3 Baseline Characteristics

From the baseline characteristics (Table 1), a total of 2,887 participants were included in the analysis. The majority were female (70.1%) and married (58.4%). The largest age groups were 30–39 years (33.9%) and 40–49 years (32.4%). Most participants had suppressed viral load (<1000 copies/ml) (95%) while only 5% had unsuppressed viral load (≥ 1000 copies/ml).

There was no statistically significant difference in the cross-sectional distribution of sex (Pearson χ^2 $p \approx 0.54$). Unsuppressed viral load was significantly more common among participants who experienced the event (Pearson χ^2 $p \approx 0.032$). The corresponding Log-rank tests that evaluate differences across the entire survival curves are reported in Section 3.2 (sex $p = 0.609$; age group $p = 0.635$; viral load $p < 0.001$; marital status $p = 0.028$). The median follow-up time was 36 months.

3.4 Univariate and Multivariate Cox Regression

In the univariate Cox regression (Table 2), each variable was examined individually. Age group, sex, and marital status showed no statistically significant association with treatment interruption or death. Participants aged 30–39, 40–49 years, and ≥ 50 years had crude HRs close to 1 when compared with those aged ≤ 29 , and none of these associations reached significance. Similarly, male participants had a slightly elevated but non-significant hazard compared with females (Crude HR = 1.05; 95% confidence interval (CI): 0.88–1.24; $p = 0.611$).

In contrast, viral load status emerged as a strong predictor. Participants with unsuppressed viral load (≥ 1000 copies/ml) had nearly twice the hazard of treatment interruption or death compared with those who had suppressed viral load (Crude HR = 1.98; 95% CI: 1.44–2.73; $p < 0.001$).

All variables were subsequently entered into a multivariate Cox model to obtain adjusted estimates (Table 2). After controlling for age group, sex, and marital status, the association between unsuppressed viral load and treatment interruption or death remained strong and statistically significant (adjusted HR = 1.96; 95% CI: 1.42–2.70; $p < 0.001$). The HRs for age group, sex, and marital status changed only slightly and remained non-significant.

Table 2. Univariate and multivariate Cox proportional hazards regression analysis

Variable	Crude HR (95% CI)	p-Value	Adjusted HR (95% CI)	p-Value
Age group	≤29 years	1.00	1.00	–
	30–39 years	0.91 (0.70–1.17)	0.91 (0.70–1.18)	0.471
	40–49 years	0.97 (0.75–1.25)	0.96 (0.74–1.23)	0.725
	≥50 years	1.05 (0.79–1.38)	1.04 (0.79–1.37)	0.787
Sex	Female	1.00	1.00	–
	Male	1.05 (0.88–1.24)	1.04 (0.88–1.23)	0.659
Viral load status	Suppressed (<1000 copies/ml)	1.00	1.00	–
	Unsuppressed (≥1000 copies/ml)	1.98 (1.44–2.73)	1.96 (1.42–2.70)	<0.001
	Married	1.00	1.00	–
Marital status	Single	1.02 (0.85–1.21)	1.02 (0.85–1.22)	0.837
	Widowed	1.20 (0.87–1.66)	1.16 (0.83–1.61)	0.389
	Divorced	0.73 (0.46–1.15)	0.75 (0.47–1.19)	0.216
	Separated	0.98 (0.62–1.55)	1.00 (0.63–1.59)	0.996

Note: CI = confidence interval; HR = hazard ratio. The en dash (–) indicates not applicable.

3.5 Final Multivariate Model Parameters

The detailed parameters of the final multivariate Cox model are presented in Table 3.

Table 3. Final multivariate Cox proportional hazards model

Parameter	Coefficient (β)	Std. Error	Z	p-Value	95% CI of β	HR (e ^β)	95% CI of HR
Age 30–39 years (vs. ≤29)	–0.0943	0.1308	–0.72	0.471	–0.3507 to 0.1621	0.910	0.704–1.176
Age 40–49 years (vs. ≤29)	–0.0458	0.1301	–0.35	0.725	–0.3008 to 0.2093	0.955	0.740–1.233
Age ≥50 years (vs. ≤29)	0.0382	0.1415	0.27	0.787	–0.2391 to 0.3154	1.039	0.787–1.371
Male (vs. Female)	0.0387	0.0877	0.44	0.659	–0.1333 to 0.2106	1.039	0.875–1.234
Unsuppressed VL (≥1000)	0.6726	0.1627	4.13	<0.001	0.3537 to 0.9916	1.959	1.424–2.696
Single (vs. Married)	0.0189	0.0922	0.21	0.837	–0.1618 to 0.1997	1.019	0.851–1.221
Widowed (vs. Married)	0.1457	0.1692	0.86	0.389	–0.1859 to 0.4772	1.157	0.830–1.612
Divorced (vs. Married)	–0.2917	0.2358	–1.24	0.216	–0.7539 to 0.1704	0.747	0.471–1.186
Separated (vs. Married)	–0.0011	0.2360	–0.00	0.996	–0.4636 to 0.4614	0.999	0.629–1.586

Note: VL = viral load; HR = hazard ratio; CI = confidence interval.

Based on these parameters, the model can be expressed as:

$$h(t) = h_0(t)e^{\left[(-0.0943(Age_{30-39}) - 0.0458(Age_{40-49}) + 0.0382(Age_{\geq 50}) + 0.0387(Male) + 0.6726(Unsuppressed\ VL) + 0.0189(Single) + 0.1457(Widowed) - 0.2917(Divorced) - 0.0011(Separated))\right]} \quad (4)$$

where, the coefficient for unsuppressed viral load ($\beta = 0.6726$) was the only statistically significant parameter ($p < 0.001$). Exponentiating this coefficient yielded an adjusted HR of 1.96, confirming that participants with unsuppressed viral load had a 96% higher hazard of the event compared with those with suppressed viral load, after adjustment for all other variables in the model.

3.6 Discussion of the Findings

This study examined factors associated with treatment interruption or death among HIV patients receiving care on ART at Specialist Hospital Yola, Adamawa State. The most important finding was that unsuppressed viral load (≥1000 copies/ml) was a strong and independent predictor of treatment interruption or death. After adjusting for

age group, sex, and marital status, HIV patients on ART with unsuppressed viral load had nearly twice the hazard of treatment interruption or death compared with those who had suppressed viral load (adjusted HR = 1.96; 95% CI: 1.42–2.70; $p < 0.001$).

This finding is consistent with evidence from large Nigerian cohorts of HIV patients on ART. Chun et al. (2022), in a retrospective longitudinal study involving over 400,000 HIV patients on ART across 18 Nigerian states, demonstrated that detectable viraemia significantly increased the risk of subsequent virological non-suppression and failure. Similarly, Tomescu et al. (2023), in a large cross-sectional analysis of 585,632 HIV patients on ART in Nigeria, reported that viral non-suppression (≥ 1000 copies/ml) remained an important programmatic challenge, with several clinical and demographic factors influencing its occurrence.

In the present study, socio-demographic variables (age group, sex, and marital status) were not significantly associated with the outcome in either univariate or multivariate models. This contrasts findings by Tomescu et al. (2023), who reported higher likelihood of viral non-suppression among males and younger age groups in Nigeria. A study conducted at the same facility (Specialist Hospital Yola) by Stephen et al. (2024) found an adherence rate of 88.4% among HIV patients on ART, with forgetfulness being the most common reason for missed doses, thus highlighting the importance of adherence support in this setting.

The Kaplan-Meier curves in Figure 3 showed a clear separation between participants with suppressed and unsuppressed viral load. Participants with unsuppressed viral load (≥ 1000 copies/ml) had significantly poorer survival experience compared with those with suppressed viral load (Log-rank test $p < 0.001$). Although the Log-rank test for marital status reached statistical significance, the practical differences between the major categories were limited, to be consistent with the non-significant adjusted HRs for marital status.

The proportional hazards assumption was satisfied, as confirmed by Schoenfeld residual plots (Figure 1), log-minus-log plots (Figure 2), and the global Schoenfeld test ($p = 0.669$). This supports the validity of the Cox model used in the analysis.

3.7 Limitations

This study has several limitations. It was a single-centre retrospective analysis based on routine hospital records, so residual confounding could not be eradicated. Important clinical covariates such as CD4 cell count, ART regimen, adherence measures, nutritional status, and co-infections were not available or were incompletely recorded, thus not being able to be included in the models. In addition, the quality and completeness of routine programme data may have introduced some misclassification of event or censoring status. These limitations should be considered when interpreting the findings. Future prospective multi-centre studies with richer clinical data are recommended.

4. Conclusions

This study among HIV patients on antiretroviral therapy at Specialist Hospital Yola demonstrated that unsuppressed viral load (≥ 1000 copies/ml) was the only independent predictor of treatment interruption or death. Patients with unsuppressed viral load had nearly twice the hazard of experiencing treatment interruption or death compared with those who had suppressed viral load, after adjusting for age group, sex, and marital status. Socio-demographic factors such as age, sex, and marital status were not significantly associated with the outcome.

These findings highlighted the critical importance of achieving and sustaining viral suppression among HIV patients on ART. Strengthening routine viral load monitoring, adherence counselling, and prompt clinical management of patients with unsuppressed viral load should be prioritized at Specialist Hospital Yola and similar treatment centres to improve patient treatment outcomes and to attain the broader goal of HIV epidemic control in Nigeria.

Author Contributions

Conceptualisation, E.R. and G.A.S.; methodology, E.R. and G.A.S.; software, E.R.; validation, E.R., G.A.S. and M.S.Y.; formal analysis, E.R.; investigation, E.R. and R.S.H.; resources, E.R.; data curation, E.R.; writing original draft preparation, E.R.; writing—review and editing, R.E., G.A.S., M.S.Y. and R.S.H.; visualization, E.R.; supervision, E.R.; project administration, E.R. All authors have read and agreed to the published version of the manuscript.

Ethical Approval

The present work was a secondary analysis of fully anonymized routine clinical records. No additional patient contact or collection of identifiable information was involved. The study was conducted in accordance with the ethical principles of the Declaration of Helsinki.

Data Availability

The data used to support the research findings are available from the corresponding author upon reasonable request. The data are under privacy and ethical restrictions because they are derived from routine clinical records.

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Conflicts of Interest

The authors declare no conflicts of interest.

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